TECHNICAL DOCUMENTATION

E-Commerce Supply Chain Performance Analytics Dashboard

Milestone 1

Data Modelling & KPI Foundation

1. Project Overview

The “E-Commerce Supply Chain Performance Analytics Dashboard” is a Business Intelligence project developed in Microsoft Power BI Desktop to convert raw, transaction-level e-commerce and logistics data into a structured, decision-ready reporting layer. The project uses the Olist Brazilian e-commerce public dataset — a set of relational CSV files describing orders, order items, products, customers, sellers, payments, reviews and product-category translations — as a stand-in for a real-world multi-vendor marketplace's operational data warehouse extract.Supply chain and order-fulfilment analytics is a core discipline within Business Analytics because operational performance — order volume, delivery timeliness, freight cost, and customer satisfaction — directly affects profitability and customer retention.

Raw transactional exports, however, are not analysis-ready: they are split across multiple files, contain inconsistent data types, embed free-text fields that are not useful for aggregate reporting, and have no shared, pre-built date calendar. Before any KPI can be trusted, the underlying data must be imported, cleaned, validated, and organised into a model that a reporting tool can query efficiently.

The overall workflow follows the standard BI development path: raw data is imported into Power Query, cleaned and shaped, loaded into the Power BI data model, connected through relationships, extended with a dedicated date table, and then summarised through a layer of reusable DAX measures. Those measures are surfaced as KPI cards and charts on an interactive report canvas, where slicers allow a user to filter the whole page by date, region, product category, and order status without writing any additional logic.

2. Project Objectives

Milestone 1 was guided by the following technical objectives:

1. Import the raw Olist e-commerce/logistics datasets into Power BI Desktop.
2. Clean and transform the data using Power Query (data types, column names, unwanted columns).
3. Validate data types and overall data quality for each source table.
4. Establish relationships between the imported tables inside the Power BI data model.
5. Develop a structured, reporting-oriented data model around the order and order-item grain.
6. Create a dedicated Date dimension (Dim_Date) to support time-based analysis.
7. Develop a set of core, reusable DAX measures for supply-chain KPIs.
8. Translate those measures into a defined supply-chain KPI layer.
9. Build an initial interactive dashboard using the KPI layer.
10. Review the dashboard's calculations and cross-filter interactions against the underlying data.

3. Project Milestone Roadmap

The E-Commerce Supply Chain Performance Analytics Dashboard is planned as a four-milestone project. This document reports on Milestone 1 only, Milestones 2–4 are listed here for context and are not implemented or documented in detail in this report.

Milestone	Focus	Status
Milestone 1	Data Modelling & KPI Foundation – import, cleaning, data model, Date dimension, core DAX measures, and the first interactive dashboard.	Covered in this document
Milestone 2	Inventory & Delivery Analytics – deeper analysis of delivery-time distributions, delay patterns, and stock/inventory-style tracking where source data permits.	Covered in this document

Milestone	Focus	Status
Milestone 3	Supplier & Transportation Analytics – seller-level and logistics-route performance, freight efficiency, and carrier/transport analysis.	Covered in this document
Milestone 4	Warehouse Analytics & Final Dashboard – consolidation of all milestones into a final, warehouse-oriented executive dashboard.	not started

4. Dataset Description

Nine CSV files were provided for this project. Eight of them were imported into the Power BI data model as query tables; the geolocation file was supplied alongside the other datasets but was not loaded into the Milestone 1 data model .

Table 4.1 – Source Datasets

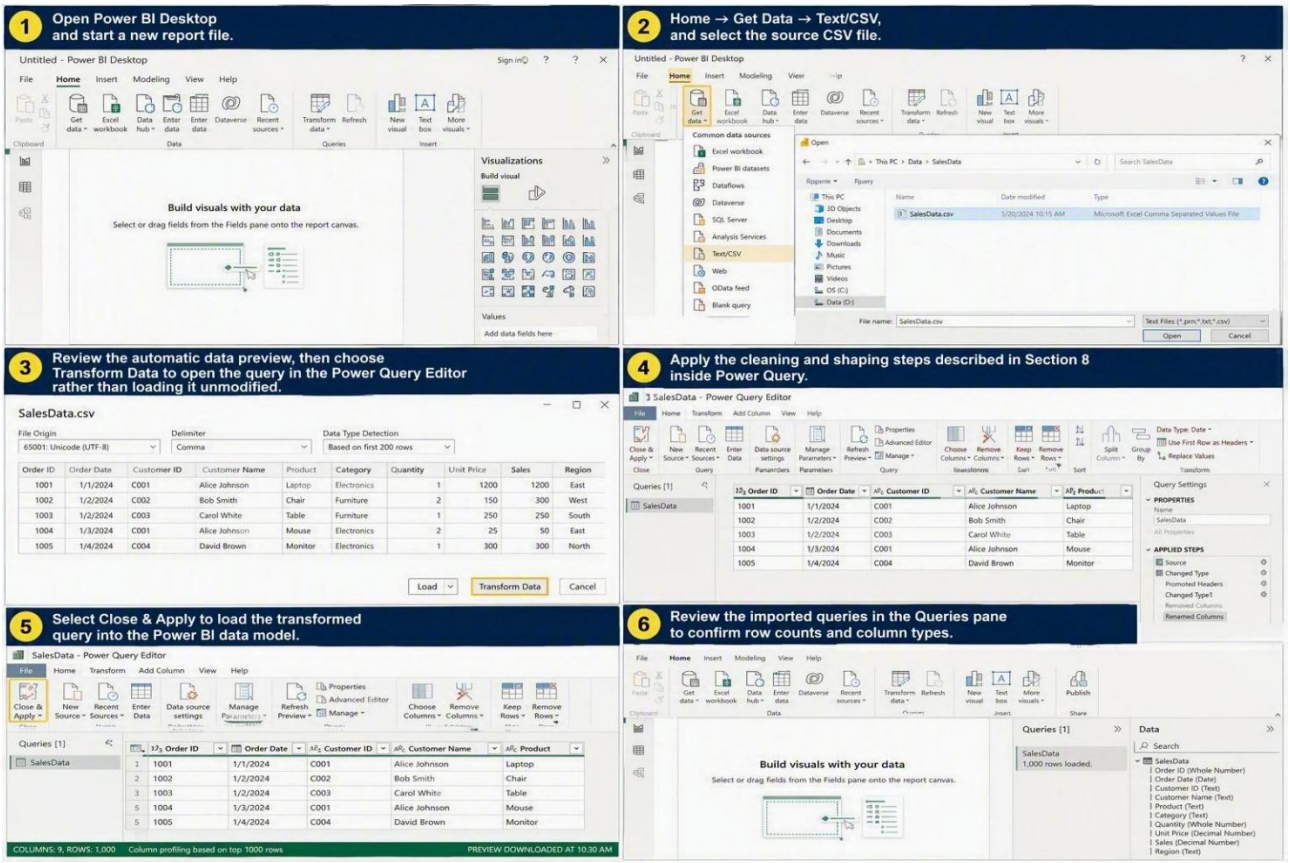
Dataset / Table	Purpose	Key Fields	Analytical Use
olist_orders_dataset (99,441 rows)	One row per order; the order-level fact table and the hub the other tables connect through.	order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date	Order status mix, delivery-time calculations, order dating/trend analysis
olist_order_items_dataset (112,650 rows)	One row per product line item within an order; the line-item grain fact table used for sales and freight measures.	order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value	Total sales, total freight cost, average order value, items sold
olist_order_payments_dataset (103,886 rows)	One row per payment transaction/instalment against an order.	order_id, payment_sequential, payment_type, payment_installments, payment_value	Available for future payment-method / instalment analysis (not yet used in a Milestone 1 measure)
olist_order_reviews_dataset (99,224 rows)	One row per customer review submitted against an order.	review_id, order_id, review_score, review_creation_date, review_answer_timestamp	Average Review Score (customer satisfaction KPI)
olist_customers_dataset (99,441 rows)	One row per customer/order location record; the customer dimension.	customer_id, customer_unique_id, customer_zip_code_prefix, customer_city, customer_state	Regional (state-level) analysis of orders
olist_products_dataset (32,951 rows)	One row per product; the product dimension.	product_id, product_category_name, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm	Product-category analysis; product-level attributes for future logistics KPIs

Dataset / Table	Purpose	Key Fields	Analytical Use
olist_sellers_dataset (3,095 rows)	One row per seller/vendor; the seller dimension.	seller_id, seller_zip_code_prefix, seller_city, seller_state	Available for future seller/supplier-performance analysis
product_category_name_translation (71 rows)	Look-up table translating Portuguese product-category names to English.	product_category_name, product_category_name_english	Supplies the English category labels used in the Sales by Product Category chart
olist_geolocation_dataset (1,000,163 rows)	Zip-code-prefix to latitude/longitude look-up.	geolocation_zip_code_prefix, geolocation_lat, geolocation_lng, geolocation_city, geolocation_state	Uploaded with the project but not imported into the Milestone 1 model – reserved for a future mapping visual

5. Data Import Process

Each of the eight source tables was brought into the Power BI data model as an individual Power Query CSV import, following the same repeatable sequence:

1. Open Power BI Desktop and start a new report file.
2. Home → Get Data → Text/CSV, and select the source CSV file.
3. Review the automatic data preview, then choose Transform Data to open the query in the Power Query Editor rather than loading it unmodified.
4. Apply the cleaning and shaping steps described in Section 8 inside Power Query.
5. Select Close & Apply to load the transformed query into the Power BI data model.
6. Review the imported queries in the Queries pane to confirm row counts and column types.



6. Power Query – Data Cleaning and Transformation

Given below the transformations that are present in each query's Applied Steps, grouped by the type of operation. Not every table required every category of transformation. where a category was not used for a given table, this is stated explicitly rather than assumed.

6.1 Data Type Correction

- Assigned correct data types to all columns after promoting headers.
- Text fields (IDs, statuses) kept as **Text**.
- Dates converted to **Date/DateTime**.
- Numeric fields converted to **Decimal Number** or **Whole Number**.
- Ensures accurate relationships, calculations, and DAX measures.

6.2 Column Renaming

- Corrected misspelled product columns:
  - product_name_lenght → Product_Name_Length
  - product_description_lenght → Product_Description_Length
- Renamed generic translation table columns (Column1, Column2) to meaningful names.
- Improves readability and DAX usability.

6.3 Missing Value Handling

- No null values were replaced.
- Blank values were intentionally preserved because they represent valid business situations (e.g., undelivered or cancelled orders).
- Prevents KPI distortion caused by artificial replacements.

6.4 Removing Unnecessary Columns

- Removed:
  - review_comment_title
  - review_comment_message
- These text fields contained many blanks and were not required for KPI calculations.
- Reduced model size and improved performance.

6.5 Duplicate Handling

- No duplicate-removal step was required.
- All imported tables contained zero fully duplicated rows.
- Duplicate checks were completed during data validation.

6.6 Date and Time Preparation

A calculated column, Order_Purchase_Date, was added to olist_orders_dataset that strips the time component from order_purchase_timestamp.

```
Order_Purchase_Date =
DATE(
    YEAR(olist_Orders_dataset[order_purchase_timestamp]),
    MONTH(olist_Orders_dataset[order_purchase_timestamp]),
    DAY(olist_Orders_dataset[order_purchase_timestamp])
)
```

- Created Order_Purchase_Date from order_purchase_timestamp.
- Removed time component for easier date analysis and calendar relationships.
- Implemented as a DAX calculated column.

6.7 Product Category Preparation

- Cleaned category names using Text.Clean.
- Removed rows containing category-name errors.
- Produced a clean Portuguese-to-English category lookup table.

6.8 Data Validation

Validated data through:

- Type conversion checks
- Row count verification
- Null value review
- Duplicate checks

Ensured data quality before creating relationships and measures.

7. Data Model Design

The Milestone 1 data model is organised around olist_orders_dataset as the central, order-grain table, with olist_order_items_dataset acting as the transaction-line fact table that carries the price and freight_value measures used by most of the KPI layer. The following seven relationships exist in the model:

From Table → Column	To Table → Column	Cardinality	Filter Direction
olist_order_items_dataset[order_id]	olist_orders_dataset[order_id]	Many : 1	Single (Order Items → Orders)
olist_order_payments_dataset[order_id]	olist_orders_dataset[order_id]	Many : 1	Single (Payments → Orders)
olist_order_reviews_dataset[order_id]	olist_orders_dataset[order_id]	Many : 1	Single (Reviews → Orders)
olist_customers_dataset[customer_id]	olist_orders_dataset[customer_id]	1 : 1	Both
olist_order_items_dataset[product_id]	olist_products_dataset[product_id]	Many : 1	Single (Order Items → Products)
olist_order_items_dataset[seller_id]	olist_sellers_dataset[seller_id]	Many : 1	Single (Order Items → Sellers)
olist_products_dataset[product_category_name]	product_category_name_translation[product_category_name]	Many : 1	Single (Products → Category Translation)



The data model was designed using a star-schema approach, with olist_orders_dataset serving as the central table connected to customers, payments, reviews, order items, and the date dimension. The olist_order_items_dataset acts as a bridge between orders, products, and sellers, enabling detailed sales and product analysis. A separate Dim_Date table provides support for time-based reporting, while the product_category_name_translation table links product categories to their English descriptions. The relationships are primarily configured as one-to-many connections, ensuring efficient filtering, accurate aggregations, and reliable KPI calculations across sales, customer, product, seller, payment, review, and date-related analyses.

8. Date Dimension

A dedicated Date table, Dim_Date, is required because transaction dates are spread across several tables and several different fields (order purchase, approval, carrier hand-off, customer delivery, estimated delivery, review dates), each with its own range. Without a single shared calendar, time-based slicers and Year/Month visuals would either have to reference one specific transaction date field directly or would produce inconsistent results across visuals that use different date columns.

Table:- Dim_Date

```
Dim_Date =
CALENDAR(
    MIN(olist_orders_dataset[order_purchase_timestamp]),
    MAX(olist_orders_dataset[order_purchase_timestamp]))
```

The CALENDAR() function generates one row per calendar day between the earliest and latest order_purchase_timestamp values in olist_orders_dataset — the source data spans 4 September 2016 to 17 October 2018. Four calculated columns extend the base Date column:

Column	DAX Expression	Purpose
Year	YEAR(Dim_Date[Date])	Whole-number year, used for year-level filtering and sorting
Month Number	MONTH(Dim_Date[Date])	Whole-number month (1–12), used to sort month names chronologically
Year Month	FORMAT(Dim_Date[Date], "YYYY-MM")	Sortable year-month label for trend axes
Quarter	"Q" & FORMAT(Dim_Date[Date], "Q")	Quarter label (e.g., Q1–Q4) for quarterly grouping

9. DAX Development

DAX (Data Analysis Expressions) is the formula language used in Power BI to create calculated columns, tables, and measures. In this project, KPI calculations were primarily implemented using measures because they dynamically respond to filters, slicers, and report interactions, ensuring accurate and context-aware business insights.

Total Sales

Home table: olist_order_items_dataset

```
Total Sales = SUM(olist_order_items_dataset[price])
```

Total Freight Cost

Home table: olist_order_items_dataset

```
Total Freight Cost = SUM(olist_order_items_dataset[freight_value])
```

Total Orders

Home table: olist_order_items_dataset

```
Total Orders = DISTINCTCOUNT(olist_order_items_dataset[order_id])
```

Total Logistics Value

Home table: olist_order_items_dataset

```
Total Logistics Value = [Total Sales] + [Total Freight Cost]
```

Average Order Value

Home table: olist_order_items_dataset

```
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders]
)
```

Total Items Sold

Home table: olist_order_items_dataset

```
Total Items Sold = COUNTROWS(olist_order_items_dataset)
```

Average Review Score

Home table: olist_order_items_dataset

```
Average Review Score = AVERAGE(olist_order_reviews_dataset[review_score])
```

Average Freight per Order

Home table: Dim_Date

```
Average Freight per Order =
DIVIDE(
    [Total Freight Cost],
    [Total Orders]
)
```

Delivered Orders

Home table: Dim_Date

```
Delivered Orders =
CALCULATE(
    [Total Orders],
    olist_orders_dataset[order_status] = "delivered"
)
```

Cancelled Orders

Home table: Dim_Date

```
Cancelled Orders =
CALCULATE(
    [Total Orders],
    olist_orders_dataset[order_status] = "canceled"
)
```

Cancelled order %

Home table: Dim_Date

```
Cancelled order % = DIVIDE([Cancelled Orders],[Total Orders])*100
```

Average Delivery Days

Home table: Dim_Date

```
Average Delivery Days =
AVERAGEX(
    FILTER(
        olist_orders_dataset,
        NOT ISBLANK(olist_orders_dataset[order_delivered_customer_date])
    ),
    DATEDIFF(
        olist_orders_dataset[order_purchase_timestamp],
        olist_orders_dataset[order_delivered_customer_date],
        DAY
    )
)
```

Delayed Orders

Home table: Dim_Date

```
Delayed Orders =
CALCULATE(
    [Total Orders],
    FILTER(
        olist_orders_dataset,
        NOT ISBLANK(olist_orders_dataset[order_delivered_customer_date])
            &&
        olist_orders_dataset[order_delivered_customer_date]
            > olist_orders_dataset[order_estimated_delivery_date]
    )
)
```

Delayed Order %

Home table: Dim_Date

```
Delayed Order % = DIVIDE([Delayed Orders],[Delivered Orders])*100
```

10. KPI Foundation

The six measures below are the ones placed on report visuals in Milestone 1.

KPI	DAX Measure	Business Meaning	Decision Use
Total Orders	[Total Orders]	Number of distinct orders in the filtered period/region/category	Tracks overall demand volume
Total Sales	[Total Sales]	Gross product revenue in the filtered period/region/category	Tracks top-line revenue performance
Total Freight Cost	[Total Freight Cost]	Total shipping cost incurred	Tracks logistics cost exposure
Average Order Value	[Average Order Value]	Average revenue per order	Indicates basket-size trends
Average Delivery Days	[Average Delivery Days]	Average time from purchase to customer delivery	Core delivery-speed / fulfilment-reliability indicator
Customer Rating	[Average Review Score]	Average post-purchase review score (1–5)	Tracks customer satisfaction alongside operational metrics

11. KPI Card Design

Six KPI cards are used in the dashboard are given below:

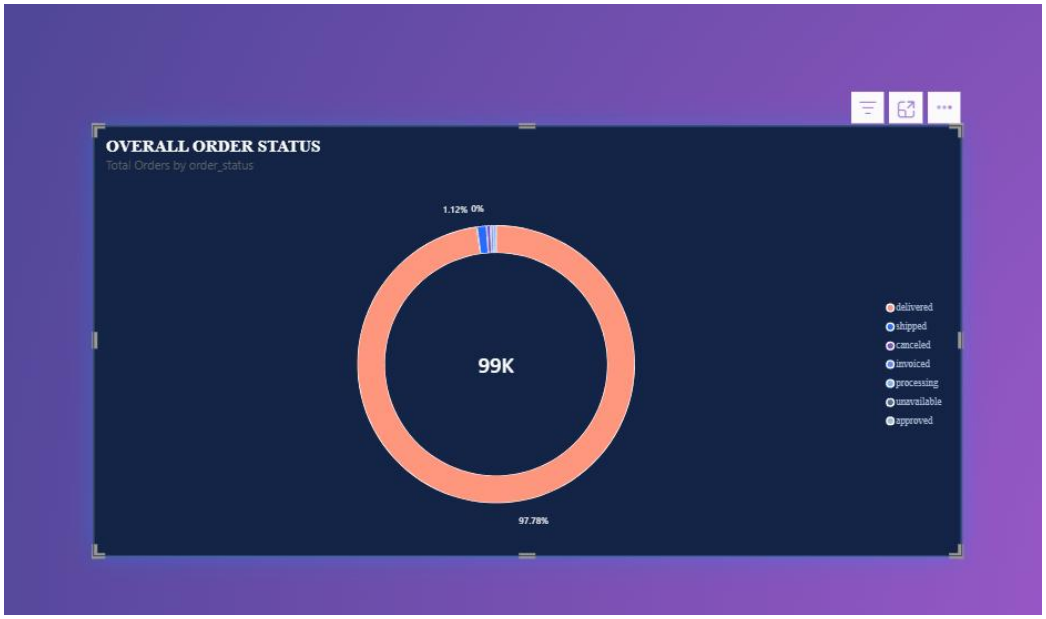
1. Total Orders – bound to [Total Orders]
2. Total Sales – bound to [Total Sales]
3. Total Freight Cost – bound to [Total Freight Cost]
4. Average Order Value – bound to [Average Order Value]
5. Average Delivery Days – bound to [Average Delivery Days]
6. Customer Rating – bound to [Average Review Score]



Figure 11.1: Milestone 1 KPI Card Section

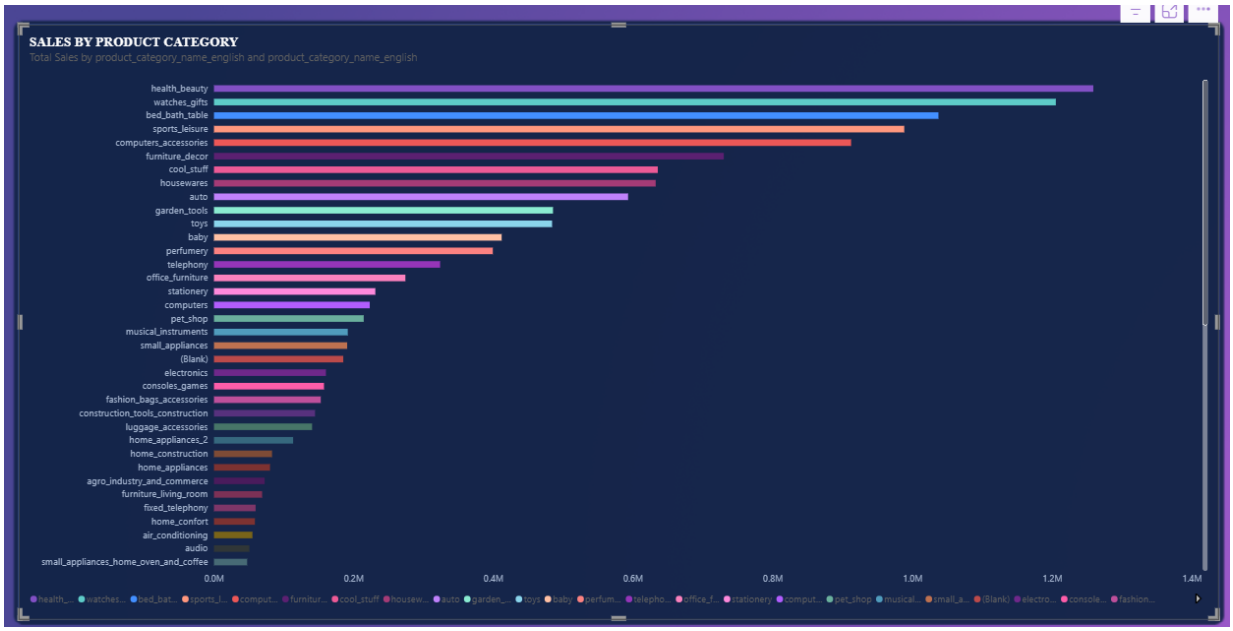
12. Order Status Visualization

The **Overall Order Status** visual uses a **Donut Chart** to display the distribution of orders across different fulfilment statuses. By plotting **order_status** as the legend and **[Total Orders]** as the value, the chart clearly shows the proportion of delivered, cancelled, shipped, and other order categories. This helps users quickly understand fulfilment performance and dynamically updates based on report filters and slicer selections.



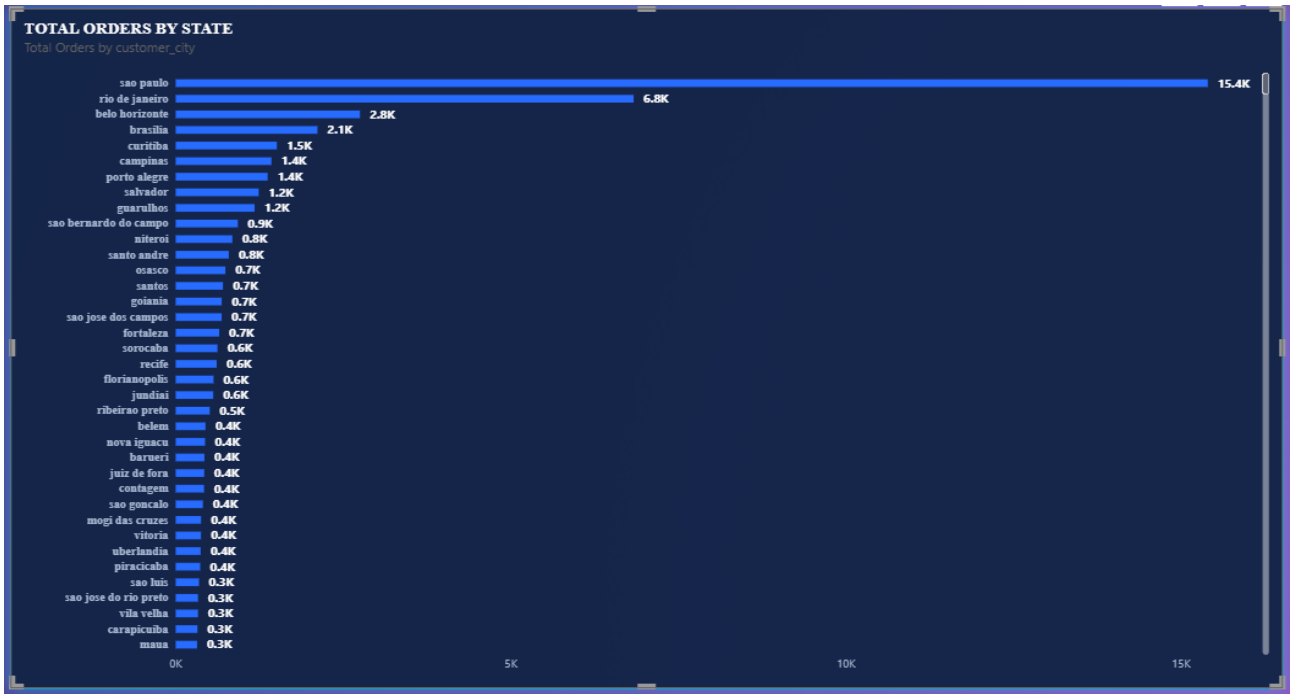
13. Product / Category Analysis

A bar chart ranks product categories by total sales, letting users compare category performance at a glance and identify the highest- and lowest-revenue categories within the current filter context. Because the category field is drawn from the English translation table rather than the raw Portuguese product_category_name, the two untranslated categories identified in Section 9 will not display a label in this chart.



14. Regional Analysis

Customer orders are broken down across the 27 distinct customer_state values present in the customer data, giving a state-by-state view of where order volume is concentrated. No Top-N filter was found configured on this visual in the extracted report metadata, so it currently displays all states rather than a curated top set; this is left as-is here rather than assumed. Business interpretation: states with disproportionately high order counts relative to population or seller presence are candidates for closer logistics-cost or delivery-time review in Milestone 2/3.



15. Interactive Slicers

Four slicers are placed along the right-hand side and lower area of Page 1, each controlling a distinct filter dimension:

Slicer	Field Used	Filtering Behaviour
Order Status	order_status (olist_orders_dataset)	Restricts all visuals to the selected order status value(s)
Year	Year (Dim_Date, subject to the relationship note in Section 10)	Restricts all visuals to the selected year(s)
Month	Dim_Date Date Hierarchy – Month level	Restricts all visuals to the selected month(s) within the selected year
Customer State	customer_state (olist_customers_dataset)	Restricts all visuals to the selected state(s)



16. Dashboard Layout

The final Milestone 1 dashboard (Page 1, 1920×1080 canvas) is arranged in four horizontal bands:



This layout was chosen so that the highest-priority information — the six headline KPIs — is visible immediately below the title without scrolling, followed by a trend view and a regional view side-by-side (so "when" and "where" can be compared at a glance), and finally the category and status breakdowns at the bottom. Keeping all four slicers on the right-hand side, rather than scattering them across the canvas, gives users one consistent place to look for filtering controls regardless of which chart they are focused on.

17. Milestone 1 Achievements

Milestone 1 delivered the following, based on the data model, DAX layer and report canvas reviewed in this document:

- Eight source datasets (99,441 orders and their related items, payments, reviews, customers, products, sellers and category translations) were integrated into a single Power BI data model.
- Each source table was cleaned in Power Query — typed correctly, renamed where the source headers were inconsistent, and stripped of unusable free-text columns — with zero duplicate rows remaining across all eight loaded tables.
- A seven-relationship data model was established around olist_orders_dataset, connecting customers, order items, products, sellers, payments, reviews and category translations.
- A dedicated Dim_Date calendar table was created, spanning the full order date range with Year, Month Number, Year Month and Quarter columns.
- Fifteen DAX measures were authored, covering order volume, sales, freight cost, order value, item volume, review score, delivery time, delay rate and cancellation rate.
- A six-KPI card layer and four supporting charts (sales trend, orders by state, order status mix, sales by category) were built on a dark, corporate dashboard theme.
- Four interactive slicers (Order Status, Year, Month, Customer State) were added to let users explore the KPI layer without leaving the report canvas.
- The model, DAX logic and dashboard were reviewed against the source data (Sections 9 and 25) to identify data-quality gaps and validation notes ahead of Milestone 2.

Milestone 2: Inventory & Delivery Analytics

1. Milestone 2 Overview

- **Expanded the Olist Power BI solution** by introducing two dedicated analytical report pages: **Inventory Analysis** and **Delivery Analysis**, building on the data model and KPI framework developed in Milestone 1.
- **Developed inventory-focused dashboards** to monitor product performance, inventory activity, seller contribution, and regional trends using interactive charts and summary KPI cards.
- **Created delivery performance visuals** to analyze delivery times, customer shipment experience, and overall logistics reliability through comparison and trend-based reporting.
- **Added delivery reliability KPIs** to the overview dashboard, enabling quick monitoring of key delivery metrics and overall operational performance.
- **Provided both high-level and detailed insights** by combining KPI cards, trend charts, comparison visuals, and a detailed analytical table, supporting data-driven decision-making across inventory and delivery operations

The sections below document what was implemented and delivered under Milestone 2 only.

2. Inventory Analysis – Achievement

An “Inventory Analysis” report page was successfully implemented in Power BI, covering inventory activity at the overall, category, seller, and product levels.

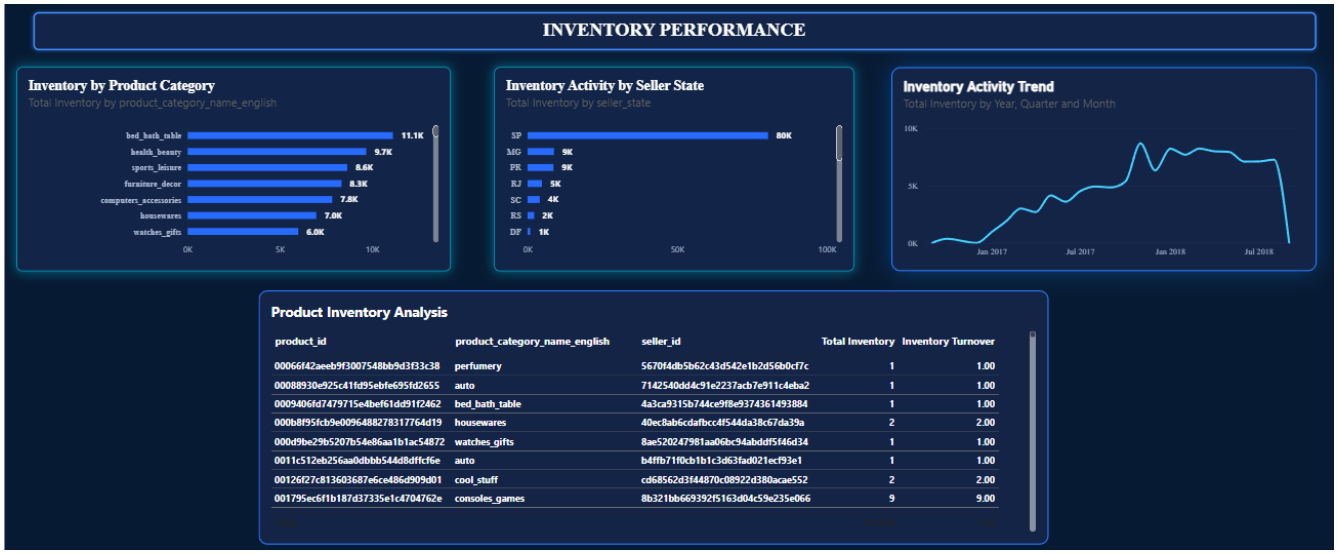
2.1 KPI Cards Implemented

KPI Card	Purpose
Total Inventory / Inventory Activity	Summarises overall order-item volume used as the inventory activity indicator.
Inventory Turnover	Indicates the relative movement rate of items.
Total Products	Count of distinct products represented in the dataset.
Slow-Moving Products	Count of products with comparatively low order-item activity.



2.2 Charts Implemented

- Product Category vs Inventory Activity – a clustered bar chart comparing inventory activity across product categories (using the translated English category names).
- Seller State vs Inventory Activity – a clustered bar chart comparing inventory activity by seller state, enabling geographic comparison of seller-side activity.
- Inventory Activity Trend over Time – a line chart showing inventory activity across the Year / Quarter / Month date hierarchy.



2.3 Product Inventory Analysis Table

A detailed table visual was implemented with the following fields:

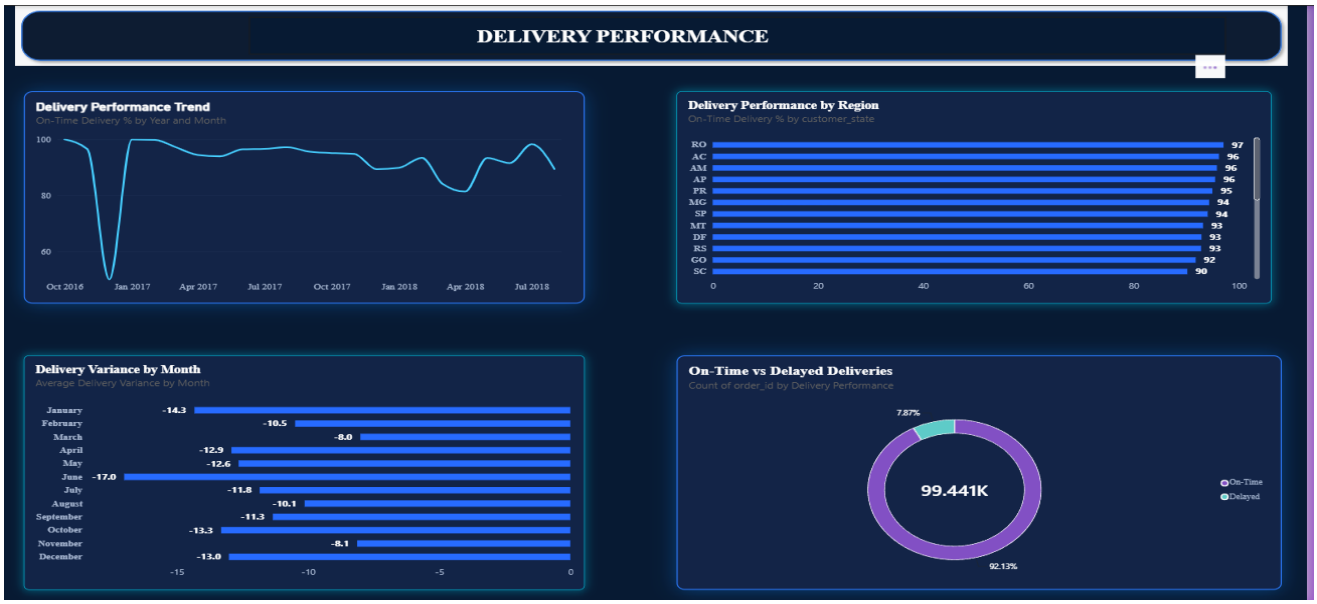
Field	Description
Product ID	Unique identifier of the product.
Product Category	English-translated product category name.
Seller ID	Unique identifier of the seller associated with the product.
Inventory Activity	Order-item volume for the product (proxy inventory measure).
Inventory Turnover / Turnover Proxy	Relative turnover indicator derived from activity volume.

3. Delivery Analysis – Achievement

A “Delivery Analysis” report page was successfully implemented, covering delivery reliability, regional comparison, trend, and variance.

3.1 KPI Cards and Visuals Implemented

Visual	Purpose
On-Time Deliveries / Delayed Deliveries	Classification of orders as on-time or delayed based on delivery date comparison.
On-Time Delivery %	Overall percentage of orders delivered on or before the estimated delivery date.
On-Time vs Delayed Deliveries (donut chart)	Visual split of delivered orders into on-time and delayed categories.
Regional Delivery Performance	Clustered bar chart comparing On-Time Delivery % across customer states.
Delivery Performance Trend over Time	Line chart of On-Time Delivery % across the Year / Month date hierarchy.
Average Delivery Variance	KPI card and monthly column chart showing the average gap between actual and estimated delivery dates.





4. Product-Level Analysis

- Inventory analysis was implemented using a **Product Category → Product ID hierarchy**.
- Users can **analyze inventory at the category level and then drill down to individual products** in both the analysis table and bar chart.
- The solution provides **hierarchical product-level analysis**, enabling comparison across categories and products, rather than a complex multi-level drill-down system.

5. Regional Analysis

- **Regional delivery analysis was implemented using Customer State**, with Customer City available for more detailed filtering when needed.
- The **Regional Delivery Performance chart compares On-Time Delivery % across states**, making it easy to identify regions with better or poorer delivery performance.
- This analysis complements the **Seller State vs Inventory Activity chart**, providing a regional perspective from both the customer and seller sides.

6. Trend Analysis

Two time-based trend visuals were successfully created for this milestone:

- Inventory Activity Trend – tracks inventory activity across the Year / Quarter / Month date hierarchy.
- Delivery Performance Trend – tracks On-Time Delivery % across the Year / Month date hierarchy, based on the delivered-customer-date.

These trend visuals allow inventory activity and delivery reliability to be observed over the reporting period rather than only as static, point-in-time totals.

7. Variance Analysis

Delivery variance analysis was successfully implemented by measuring the difference between the actual delivery date and the estimated delivery date for each order:

Variance	Interpretation
Negative variance	Order delivered earlier than the estimated delivery date.
Zero variance	Order delivered exactly on the estimated delivery date.
Positive variance	Order delivered later than the estimated delivery date (delayed).

The Average Delivery Variance KPI summarises this gap across all delivered orders, and a monthly column chart presents the variance trend so that months with wider or narrower delivery gaps can be compared.

8. Technical Implementation

This section documents the Power BI implementation completed specifically for Milestone 2, including the visuals built, the DAX measures created, filters/slicers used, and date-based and table/chart interactions configured.

8.1 Visuals and Interactions

- KPI cards for Total Inventory, Inventory Turnover, Total Products, Slow-Moving Products, On-Time Delivery %, and Average Delivery Variance.
- Clustered bar charts for Product Category vs Inventory Activity, Seller State vs Inventory Activity, and Customer State vs On-Time Delivery %.
- A donut chart for the On-Time vs Delayed Deliveries split.
- Line charts for the Inventory Activity Trend and the Delivery Performance Trend, both built on the date hierarchy (Year / Quarter / Month).
- A detailed table (Table Ex) visual for product-level inventory analysis.
- Standard cross-highlighting/cross-filtering interactions between the charts, cards, and table on each report page.

8.2 DAX Measures

The following measures were created to support Milestone 2 reporting. Names reflect the measures used in the report; expressions are shown in standard DAX form.

Total Inventory (Inventory Activity)

```
Total Inventory =
SUM ( olist_order_items_dataset[order_item_id] )
```

Sums order-item volume per product/category/seller as the proxy indicator of inventory activity, since the dataset does not provide physical stock quantities.

Inventory Turnover (Turnover Proxy)

```
Inventory Turnover =
DIVIDE (
    [Total Inventory],
    DISTINCTCOUNT ( olist_products_dataset[product_id] )
)
```

Expresses activity volume relative to the number of distinct products, used purely as a relative turnover proxy rather than a true stock-based turnover ratio.

Total Products

```
Total Products =
DISTINCTCOUNT ( olist_products_dataset[product_id] )
```

Counts the distinct products represented across the inventory analysis.

Slow-Moving Products

```
Slow-Moving Products =
CALCULATE (
    DISTINCTCOUNT ( olist_products_dataset[product_id] ),
```

```
    FILTER (
        VALUES ( olist_products_dataset[product_id] ),
        [Total Inventory] <= [Slow-Moving Threshold]
    )
)
```

Identifies products whose inventory activity falls at or below a defined low-activity threshold, flagging them as slow-moving.

On-Time Deliveries

```
On-Time Deliveries =
CALCULATE (
    COUNTROWS ( olist_orders_dataset ),
    olist_orders_dataset[order_delivered_customer_date]
        <= olist_orders_dataset[order_estimated_delivery_date]
)
```

Counts delivered orders where the actual delivery date is on or before the estimated delivery date.

Delayed Deliveries

```
Delayed Deliveries =
CALCULATE (
    COUNTROWS ( olist_orders_dataset ),
    olist_orders_dataset[order_delivered_customer_date]
        > olist_orders_dataset[order_estimated_delivery_date]
)
```

Counts delivered orders where the actual delivery date falls after the estimated delivery date.

On-Time Delivery %

```
On-Time Delivery % =
DIVIDE (
    [On-Time Deliveries],
    [On-Time Deliveries] + [Delayed Deliveries]
)
```

Expresses on-time deliveries as a percentage of all delivered orders; used on KPI cards and the regional and trend visuals.

Average Delivery Variance

```
Average Delivery Variance =
AVERAGEX (
    olist_orders_dataset,
    DATEDIFF (
        olist_orders_dataset[order_estimated_delivery_date],
        olist_orders_dataset[order_delivered_customer_date],
        DAY
    )
)
```

Calculates the average number of days between the estimated and actual delivery dates across delivered orders — negative for early deliveries, positive for delayed ones.

8.3 Filters, Slicers, and Date-Based Analysis

- Page-level filters applied to keep the inventory and delivery visuals scoped to valid, delivered orders.
- Date hierarchy (Year / Quarter / Month) used consistently across the trend visuals for both inventory activity and delivery performance.
- Category, seller-state, and customer-state fields used as chart axes to enable direct visual comparison without requiring separate slicer panels.

9. Milestone 2 Achievements Summary

✓ Successfully developed an inventory analytics dashboard in Power BI.

✓ Successfully implemented inventory activity and turnover proxy KPIs.

✓ Successfully created product-category and seller-region inventory analysis.

✓ Successfully developed delivery performance monitoring using on-time and delayed delivery metrics.

✓ Successfully implemented regional delivery performance analysis.

✓ Successfully created time-series trend analysis for inventory and delivery performance.

✓ Successfully implemented delivery variance analysis.

✓ Successfully developed product-level analytical views.

Milestone 3: Supplier and Transportation Performance Analytics

1. Summary

Milestone 3 extends the supply-chain Power BI solution into supplier and transportation analytics. The uploaded PBIX project contains two dedicated report pages: “SUPPLIER PERFORMANCE” and “TRANSPORTATION ANALYSIS”. These pages use seller, shipment, freight-cost, product, route, and delivery-performance measures to provide an interactive view of supplier activity and transportation efficiency.
The Supplier Performance page contains KPI cards for total suppliers, total shipments, transportation cost, average shipment cost, and average quality score. It also includes supplier-level shipment, quality, and transportation-cost visualizations, a seller slicer, a seller-state map, and a product-category analysis.
The Transportation Analysis page focuses on total shipments, transportation cost, average shipment cost, supplier-state cost, supplier transportation cost, total orders, on-time delivery percentage, route cost by shipping rate, and shipment volume by route. A shipping-route slicer is provided for interactive route analysis.

2. Milestone 3 Objectives

2.1 Supplier Analysis Objectives

- Measure the number of suppliers represented in the dataset.
- Analyze shipment volume by supplier.
- Compare supplier-level average quality scores.
- Measure transportation cost attributable to individual suppliers.
- Provide a supplier-focused scorecard through KPI cards and interactive charts.
- Enable seller-level filtering through the seller slicer.

2.2 Transportation Analysis Objectives

- Measure total shipment activity and total transportation cost.
- Monitor average shipment cost.
- Compare transportation cost across supplier states.
- Analyze transportation cost by supplier.
- Evaluate route cost and shipment volume.
- Monitor on-time delivery percentage alongside transportation metrics.
- Provide interactive route analysis through the shipping-route slicer.

3. Project and Dataset Context

The technical documentation is based on the structure detected in the uploaded Power BI project. The report contains two Milestone 3 pages and uses the Olist-style e-commerce/supply-chain tables visible in the Power BI model.

Relevant Table / Source	Field or Measure Used	Role in Milestone 3
olist_sellers_dataset	seller_id	Supplier identification and supplier-level analysis
olist_sellers_dataset	seller_state	Supplier-state transportation analysis and map

olist_order_items_dataset	Total Shipments	Shipment-volume KPI and supplier/route analysis
olist_order_items_dataset	Total Transportation Cost	Core transportation-cost KPI
olist_order_items_dataset	Average Shipment Cost	Average shipment-cost KPI
olist_order_items_dataset	Average Quality Score	Supplier quality comparison
olist_order_items_dataset	Total Orders	Order-volume KPI on transportation page
olist_orders_dataset	On-Time Delivery %	Transportation service-level indicator
product_category_name_translation	product_category_name_english	Product-category transportation analysis
Transportation_Data	Shipping Route	Route-level transportation analysis
Transportation_Data	Total Route Cost	Route cost analysis
Transportation_Data	Route Shipments	Route shipment-volume analysis

4. Supplier Performance Analysis

The "SUPPLIER PERFORMANCE" report page is the supplier-focused component of Milestone 3. Its analysis is centered on seller_id and seller_state, with shipment, transportation-cost, and quality measures.

4.1 Supplier KPI Foundation

KPI	Power BI Measure	Purpose
Total Supplier	olist_sellers_dataset.Total Suppliers	Counts the supplier population represented by the model.
Total Shipments	olist_order_items_dataset.Total Shipments	Measures shipment activity associated with the supplier analysis.
Transportation Cost	olist_order_items_dataset.Total Transportation Cost	Measures total transportation expenditure.
Avg Shipment Cost	olist_order_items_dataset.Average Shipment Cost	Measures average cost associated with shipments.
Avg Quality Score	olist_order_items_dataset.Average Quality Score	Provides a supplier-quality comparison metric.



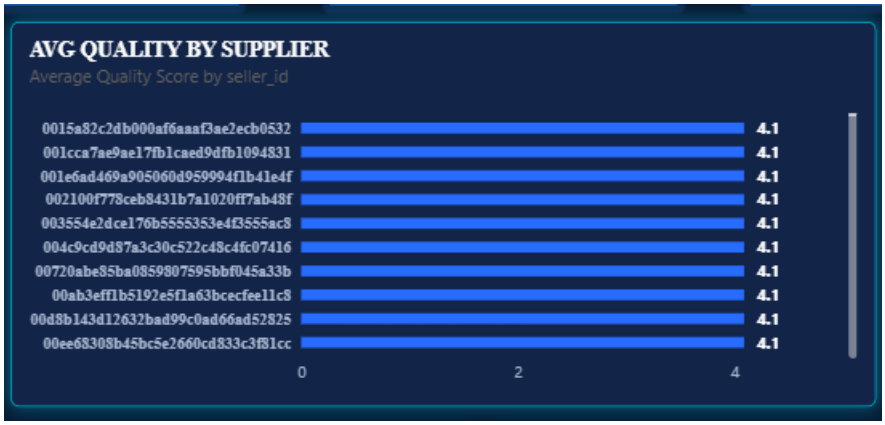
4.2 Supplier Shipment Analysis

The visual titled "SHIPMENT COUNT BY SUPPLIER" is implemented as a ribbon chart. The category is seller_id and the value is Total Shipments. The chart therefore compares supplier shipment volume and allows the report user to observe the relative position of suppliers within the visual.



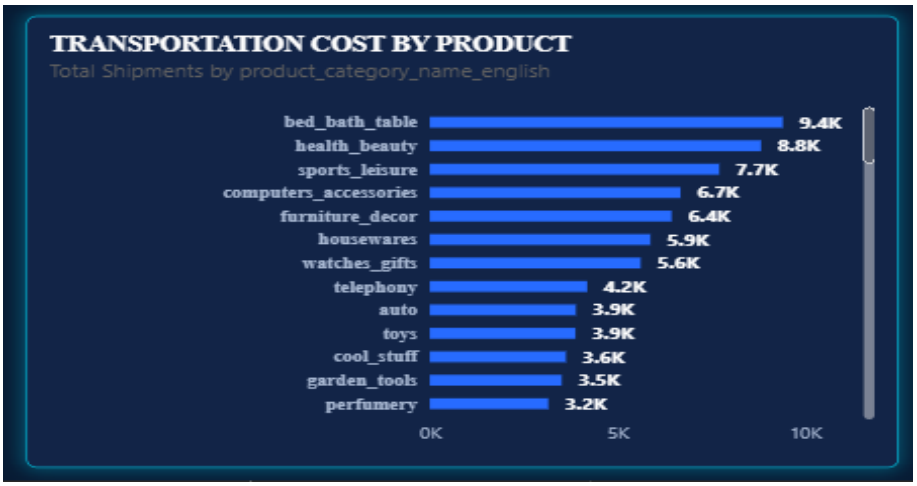
4.3 Supplier Quality Analysis

The visual titled “AVG QUALITY BY SUPPLIER” is a bar chart using seller_id as the category and Average Quality Score as the value. This visual provides a direct supplier-by-supplier comparison of the quality measure stored in the model.



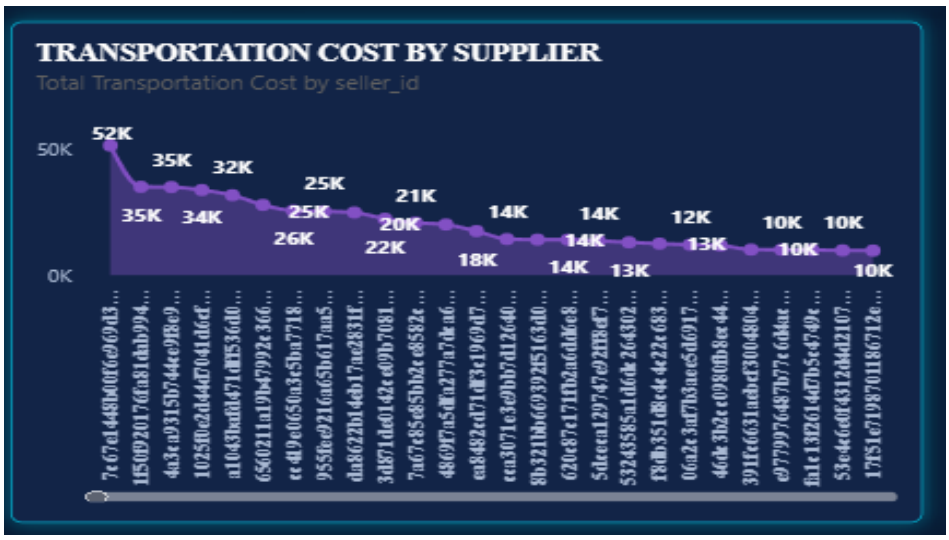
4.4 Transportation Cost by Product

The highest transportation costs are associated with bed_bath_table (9.4K) and health beauty (8.8K) products, indicating these categories drive the largest share of shipment expenses. Mid-range categories such as sports leisure and computers accessories also contribute significantly, while categories like garden tools and perfumery incur comparatively lower transportation costs.



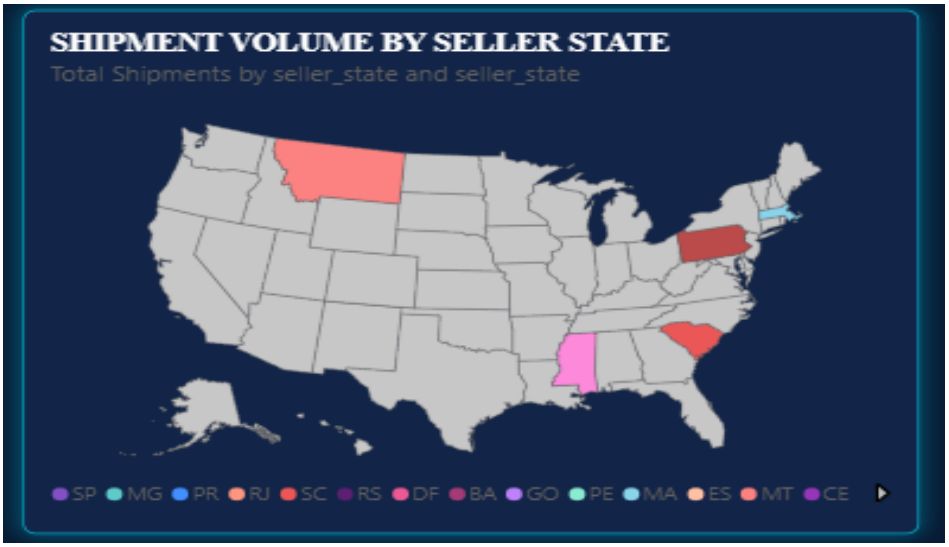
4.5 Supplier Transportation Cost Analysis

The visual titled “TRANSPORTATION COST BY SUPPLIER” is an area chart. It uses seller_id as the category and Total Transportation Cost as the value. This connects supplier activity with transportation expenditure and provides a comparative view of cost across suppliers.



4.6 Supplier Geographic Analysis

The visual titled “SHIPMENT VOLUME BY SELLER STATE” is a shape-map visual. It uses seller_state as the geographic category and Total Shipments as the value. The same seller_state field is also used as the series. This allows shipment volume to be viewed geographically by supplier state.



4.7 Supplier Filtering

A slicer titled “SELLER” is included on the Supplier Performance page and uses seller_id. The slicer provides supplier-level interactivity so that the user can isolate the analysis to selected sellers.

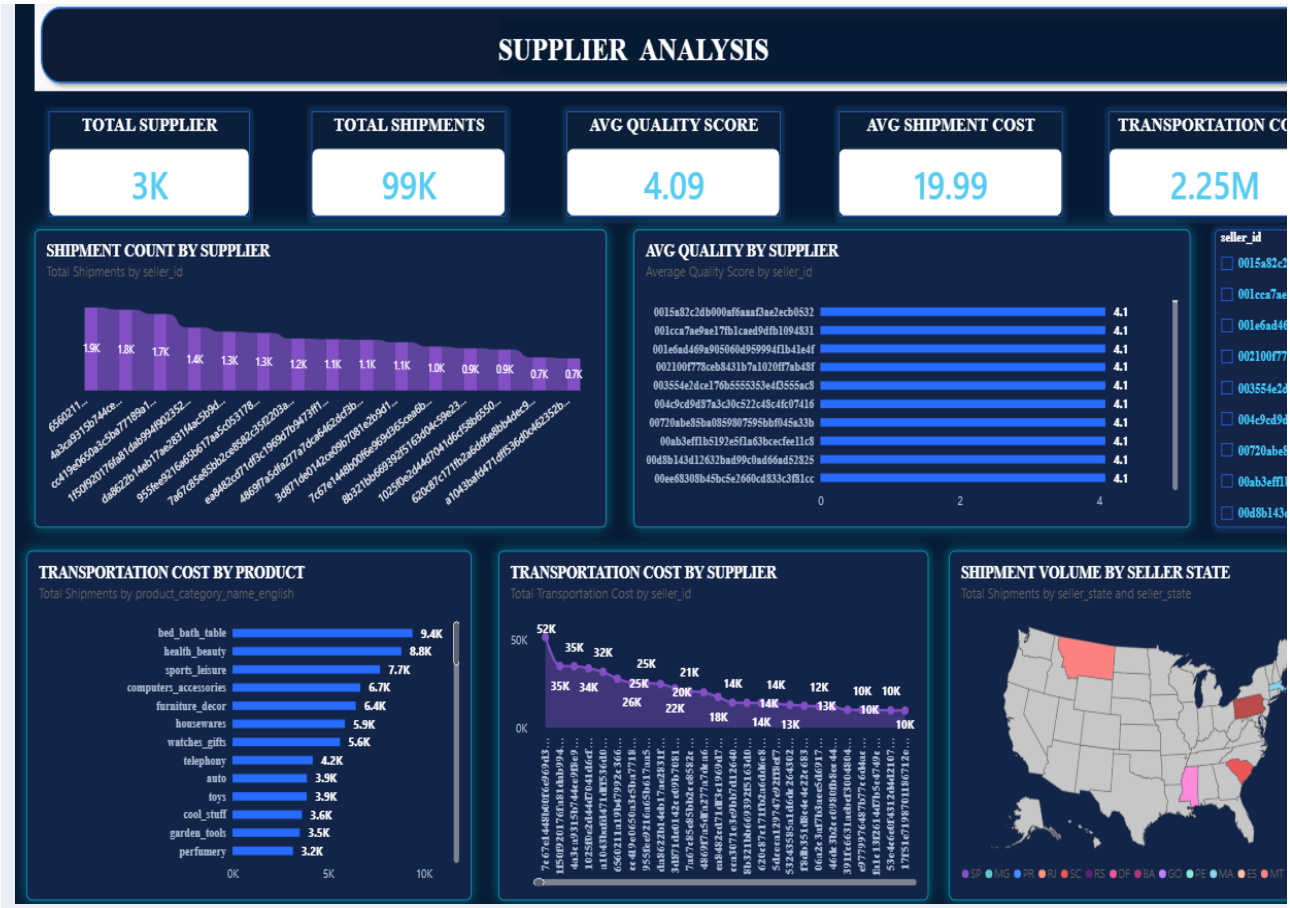


Figure 1. Supplier Performance Dashboard

5. Supplier Dashboard Visuals

Visual	Type	Category / Axis	Measure / Value
TOTAL SUPPLIER	Card	—	Total Suppliers
TOTAL SHIPMENTS	Card	—	Total Shipments
TRANSPORTATION COST	Card	—	Total Transportation Cost
AVG SHIPMENT COST	Card	—	Average Shipment Cost
AVG QUALITY SCORE	Card	—	Average Quality Score
SHIPMENT COUNT BY SUPPLIER	Ribbon chart	seller_id	Total Shipments
AVG QUALITY BY SUPPLIER	Bar chart	seller_id	Average Quality Score
TRANSPORTATION COST BY SUPPLIER	Area chart	seller_id	Total Transportation Cost
SHIPMENT VOLUME BY SELLER STATE	Shape map	seller_state	Total Shipments
SELLER	Slicer	seller_id	Interactive filter
TRANSPORTATION COST BY PRODUCT	Clustered bar chart	product_category_name_english	Total Shipments

6. Transportation Analysis

The “TRANSPORTATION ANALYSIS” page is dedicated to transportation cost, shipment volume, route performance, and delivery service-level analysis. It combines supplier geography, supplier-level cost, route-level measures, and order/delivery measures.

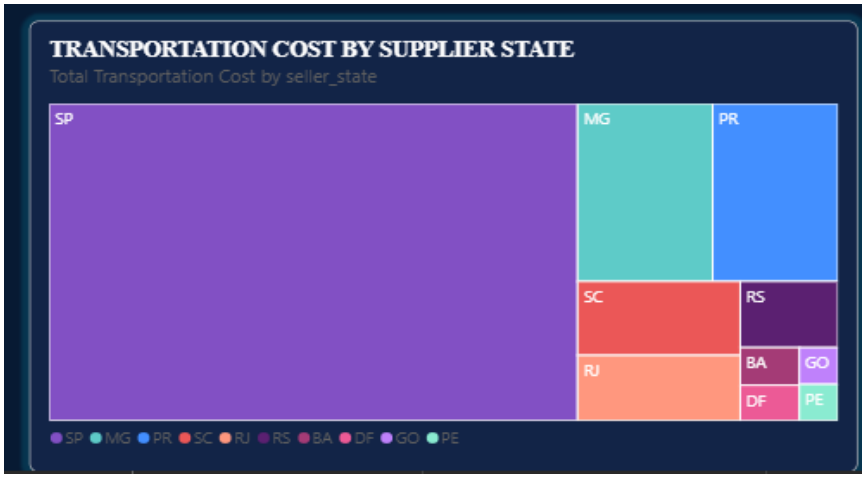
6.1 Transportation KPI Foundation

KPI	Power BI Measure	Purpose
Total Shipments	olist_order_items_dataset.Total Shipments	Measures overall shipment volume.
Transportation Cost	olist_order_items_dataset.Total Transportation Cost	Measures overall transportation expenditure.
Avg Shipment Cost	olist_order_items_dataset.Average Shipment Cost	Measures average transportation/shipment cost.
Total Orders	olist_order_items_dataset.Total Orders	Provides order-volume context for transportation analysis.
On-Time Deliver %	olist_orders_dataset.On-Time Delivery %	Measures delivery service performance.



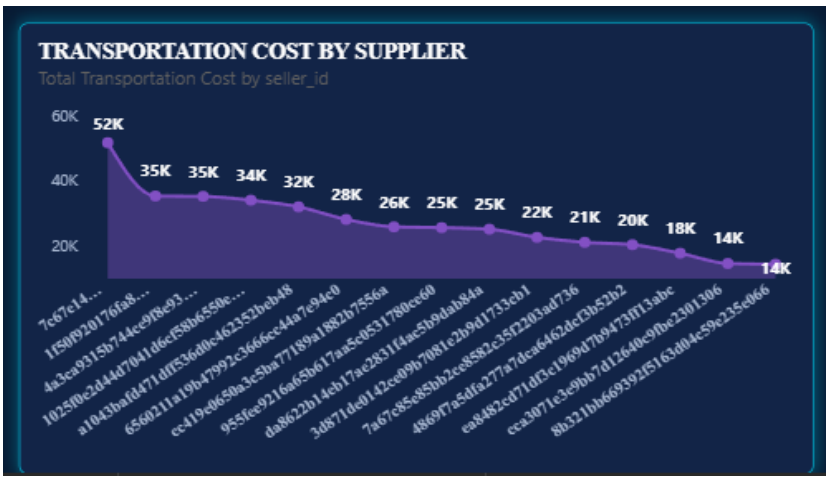
6.2 Transportation Cost by Supplier State

The “TRANSPORTATION COST BY SUPPLIER STATE” visual is a treemap. seller_state is used as the group and Total Transportation Cost is used as the value. This visual provides a compact comparison of transportation expenditure across supplier states.



6.3 Transportation Cost by Supplier

The “TRANSPORTATION COST BY SUPPLIER” visual is an area chart using seller_id as the category and Total Transportation Cost as the value. It provides supplier-level transportation-cost comparison on the dedicated transportation page.



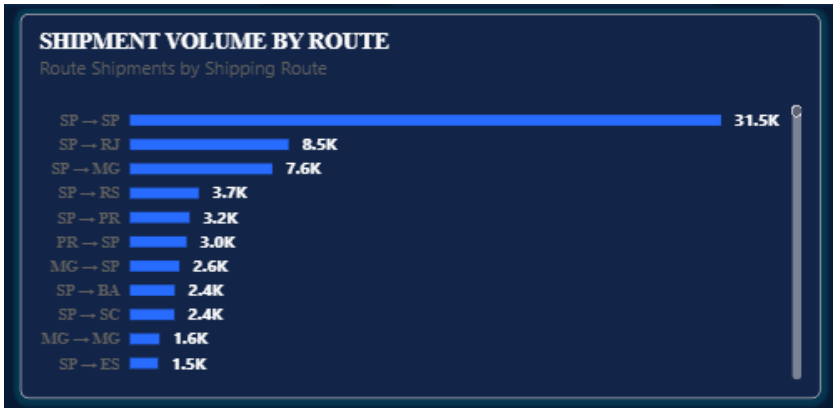
6.4 Route Cost Analysis

The “ROUTE COST BY SHIPPING RATE” visual is a line chart. The category is Shipping Route from Transportation_Data and the value is Total Route Cost. This provides route-level cost analysis using the project’s derived Transportation_Data source.



6.5 Shipment Volume by Route

The “SHIPMENT VOLUME BY ROUTE” visual is a clustered bar chart. Shipping Route is used as the category and Route Shipments as the value. This allows route-level shipment volume to be compared independently of the cost measure.



6.6 Route Filtering

A slicer titled “SELLER” is present on the Transportation Analysis page and uses Transportation_Data[Shipping Route]. Although the displayed title is SELLER, its verified field projection is Shipping Route; therefore the control functions as a route filter.

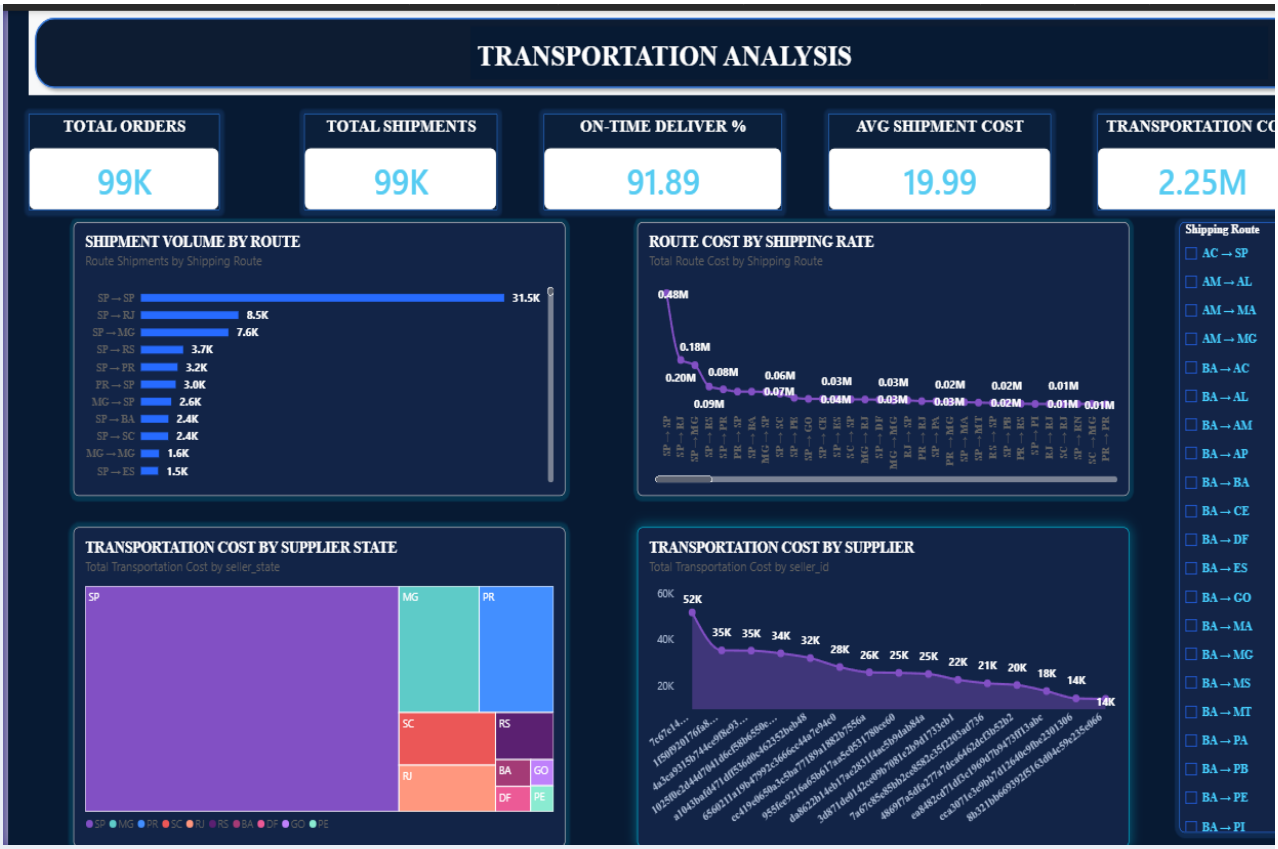


Figure 2. Transportation Analysis Dashboard

7. Transportation Dashboard Visuals

Visual	Type	Category / Group	Measure / Value
TOTAL SHIPMENTS	Card	—	Total Shipments
TRANSPORTATION COST	Card	—	Total Transportation Cost
AVG SHIPMENT COST	Card	—	Average Shipment Cost
TRANSPORTATION COST BY SUPPLIER STATE	Treemap	seller_state	Total Transportation Cost
TRANSPORTATION COST BY SUPPLIER	Area chart	seller_id	Total Transportation Cost
TOTAL ORDERS	Card	—	Total Orders
ON-TIME DELIVER %	Card	—	On-Time Delivery %
ROUTE COST BY SHIPPING RATE	Line chart	Shipping Route	Total Route Cost
SHIPMENT VOLUME BY ROUTE	Clustered bar chart	Shipping Route	Route Shipments
SELLER	Slicer	Shipping Route	Interactive route filter

8. DAX and Measure

The Milestone 3 visuals reference reusable Power BI measures rather than performing all calculations directly inside individual visuals. The verified measure names in the report layout are listed below.

Measure	Referenced By	Analytical Role
Total Suppliers	Supplier KPI card	Supplier population
Total Shipments	Supplier KPI; Transportation KPI; supplier charts	Shipment volume
Total Transportation Cost	Supplier KPI; supplier area chart; transportation visuals	Transportation expenditure
Average Shipment Cost	Supplier KPI; transportation KPI	Average shipment cost
Average Quality Score	Supplier KPI; supplier quality chart	Supplier quality comparison
Total Orders	Transportation KPI	Order-volume context
On-Time Delivery %	Transportation KPI	Delivery service level
Total Route Cost	Route cost line chart	Route-level transportation cost
Route Shipments	Route shipment bar chart	Route-level shipment volume

8.1 Total Suppliers

Total Suppliers =

DISTINCTCOUNT('olist_sellers_dataset'[seller_id])

Purpose: Counts the unique suppliers/sellers available in the dataset.

8.2 Total Shipments

Total Shipments =

COUNTROWS('olist_order_items_dataset')

Purpose: Calculates the total number of shipment/order-item records used for supplier and transportation analysis.

8.3 Total Transportation Cost

Total Transportation Cost =

SUM('olist_order_items_dataset'[freight_value])

Purpose: Calculates the total freight/transportation cost.

8.4 Average Shipment Cost

Average Shipment Cost =

AVERAGE('olist_order_items_dataset'[freight_value])

Purpose: Calculates the average freight cost associated with a shipment.

8.5 Average Quality Score

Average Quality Score =

AVERAGE('olist_order_reviews_dataset'[review_score])

Purpose: Calculates the average customer review score used as a quality indicator in supplier analysis.

8.6 Total Orders

Total Orders =

DISTINCTCOUNT('olist_orders_dataset'[order_id])

Purpose: Calculates the number of unique orders for transportation analysis.

8.7 On-Time Delivery %

On-Time Delivery % =

```
DIVIDE(
    CALCULATE(
        [Total Orders],
        'olist_orders_dataset'[delivery_status] = "On Time"
    ),
    [Total Orders],
    0
)
```

Purpose: Measures the percentage of orders delivered on time.

8.8 Total Route Cost

Total Route Cost =

SUM('Transportation_Data'[Route Cost])

Purpose: Calculates total transportation cost for each shipping route.

8.9 Route Shipments

Route Shipments =

COUNTROWS('Transportation_Data')

Purpose: Calculates the number of shipments associated with each shipping route.

9. Technical Workflow

1. Identify supplier, shipment, transportation, route, product, and delivery fields required for Milestone 3.
2. Use the existing Power BI data model established in the earlier milestones.
3. Create or reuse measures for supplier count, shipments, transportation cost, average shipment cost, quality, orders, on-time delivery, route cost, and route shipments.
4. Build the Supplier Performance page with KPI cards, supplier comparison visuals, geographic analysis, and a seller slicer.
5. Build the Transportation Analysis page with transportation KPIs, supplier-state cost analysis, supplier cost analysis, route analysis, and delivery-performance KPI.
6. Configure visual titles, formatting, axes, values, and filters.
7. Validate that each visual references the intended field or measure.
8. Test slicers and cross-visual interactions.
9. Review the final pages for consistency and usability.

10. Business Application

- Supplier monitoring: management can compare supplier shipment activity, quality score, and transportation expenditure.
- Transportation cost monitoring: total and average shipment cost provide a cost-control view.
- Geographic planning: supplier-state analysis highlights geographic concentrations of shipment activity and transportation cost.
- Route monitoring: route cost and route shipment volume provide a basis for reviewing route-level logistics activity.
- Service-level monitoring: On-Time Delivery % places delivery performance alongside transportation metrics.
- Interactive analysis: slicers allow users to move from an overall view to selected suppliers or routes.

11. Milestone 3 Achievement Summar

Milestone 3 was implemented as a dedicated Supplier Performance and Transportation Analysis module within the Power BI supply-chain dashboard. The completed pages integrate supplier-level KPIs, supplier shipment and quality analysis, transportation-cost measures, supplier-state analysis, route-level cost and shipment analysis, and an on-time delivery KPI.

Milestone 3 Requirement	Implementation in the PBIX Project
Supplier scorecards	Supplier KPI cards and supplier-level shipment, quality, and transportation-cost visuals
Transportation cost dashboard	Transportation KPI cards, supplier-state treemap, supplier cost analysis
Route performance analysis	Shipping Route cost and shipment-volume visuals
Advanced KPI visualizations	Total suppliers, shipments, transportation cost, average shipment cost, quality, orders, and on-time delivery cards

MILESTONE 4 – WAREHOUSE ANALYTICS AND FINAL DASHBOARD

4.1 Introduction

Milestone 4 focuses on the development of Warehouse Analytics and the Final Supply Chain Dashboard using Microsoft Power BI. This milestone brings together the analysis developed in the previous milestones and provides a consolidated view of supply chain performance.

The main objectives of Milestone 4 are to develop warehouse efficiency reports, create an executive-level overview dashboard, implement dashboard navigation, optimize report performance, perform final testing, and prepare the completed Power BI report for final deployment.

Milestone 4 builds upon the data modelling, inventory analysis, delivery analysis, supplier analysis, and transportation analysis completed in the earlier milestones.

4.2 Objectives of Milestone 4

The major objectives of Milestone 4 are:

1. Develop a Warehouse Efficiency Dashboard.
2. Analyse warehouse order and fulfilment performance.
3. Analyse warehouse capacity utilization and picking accuracy.
4. Analyse warehouse processing time and delivery performance.
5. Create an Executive Overview Dashboard.
6. Provide navigation between all major dashboard pages.
7. Implement warehouse drill-through analysis.
8. Optimize Power BI report performance.
9. Perform final validation and testing.
10. Prepare the final dashboard for deployment and presentation.

4.3 Warehouse Analytics

The Warehouse Efficiency Dashboard was developed to analyze warehouse-related operational performance. The dashboard provides a comparison of different proxy warehouse locations based on order volume, fulfilments performance, delivery performance, capacity utilization, picking accuracy and processing time.

The warehouse analysis includes the following key performance indicators:

- Warehouse Orders
- Warehouse Fulfilments Rate
- Warehouse On-Time Rate
- Capacity Utilization
- Picking Accuracy
- Average Processing Hours
- Warehouse Operating Cost
- Warehouse Sales

- Warehouse Freight Cost

The dashboard also includes visual analysis such as:

- Orders by Warehouse
- Capacity Utilization by Warehouse
- Picking Accuracy by Warehouse
- On-Time Delivery by Warehouse
- Average Processing Hours by Warehouse
- Orders by Zone
- Warehouse Performance Details

4.4 Warehouse Data Limitation and Proxy Methodology

The original Olist Brazilian E-commerce dataset does not contain a dedicated warehouse operational table. Therefore, a **proxy warehouse dataset** was developed using available seller geographic information and order fulfilments information.

Five proxy warehouse hubs were created:

- WH-Sao-Paulo
- WH-Curitiba
- WH-Belo-Horizonte
- WH-Rio
- WH-Salvador

These proxy warehouses allow warehouse-level analysis to be demonstrated within the Power BI project.

The following statement defines the methodology used:

The original Olist dataset did not contain a dedicated warehouse operational table. Therefore, a proxy warehouse dataset was developed using seller geographic information and order fulfillment data. Five proxy warehouse hubs were created to enable warehouse-level analysis. Capacity utilization, picking accuracy, processing hours and operating cost are proxy indicators derived from the available data and should not be interpreted as actual warehouse records.

Therefore, the warehouse analysis should be considered a **business simulation/proxy analysis** rather than actual warehouse operational reporting.

4.5 Executive Overview Dashboard

The Executive Overview was created as the main management dashboard of the project. It provides a summarized view of the overall supply chain performance and allows users to quickly understand important business KPIs.

The Executive Overview contains the following major KPI cards:

- Total Orders

- Fulfillment Rate
- Total Inventory
- Inventory Turnover
- On-Time Delivery
- Transportation Cost
- Average Shipment Cost
- Warehouse Operating Cost

The dashboard also contains the following visualizations:

- Monthly Order Trend
- Warehouse Delivery Performance
- Transportation Cost by Route
- Inventory by Product Category
- Orders by Product Category

The Executive Overview allows management to quickly identify trends, performance gaps and important areas requiring further analysis.

4.6 Transportation Analysis

The transportation section uses the existing **Route** field to analyze transportation costs.

Examples of routes include:

- SP → RJ
- MG → SP
- PR → BA

A Carrier field was not available in the Olist dataset. Therefore, carrier-level analysis was not created. Instead, transportation performance was analyzed using the available Route information.

This approach avoids introducing unsupported or artificial carrier information into the analysis.

4.7 Dashboard Navigation

A Page Navigator was implemented to improve the usability of the Power BI report.

The navigation structure contains:

- Executive Overview
- Inventory Analysis
- Delivery Analysis
- Supplier Analysis
- Transportation Analysis
- Warehouse Efficiency
- Warehouse Details

The Page Navigator was created using:

Insert → Buttons → Navigator → Page Navigator

Navigation buttons allow users to move easily between different analytical sections of the report.

A Home/Executive Overview button was also provided to return users to the main dashboard.

4.8 Warehouse Drill-Through

A **Warehouse Details** page was created to provide detailed analysis for an individual warehouse.

Users can select a warehouse from the Warehouse Efficiency dashboard and use the drill-through functionality to access detailed warehouse information.

The drill-through page allows users to analyze:

- Warehouse Orders
- Fulfillment Rate
- On-Time Rate
- Capacity Utilization
- Picking Accuracy
- Processing Hours
- Warehouse-related performance details

A Back button was also added to allow users to return to the previous dashboard.

4.9 Dashboard Performance Optimization

Several optimization techniques were considered during the final development of the report.

These include:

- Removing unnecessary fields where appropriate
- Avoiding duplicate DAX measures
- Reusing existing measures from previous milestones
- Limiting unnecessary visuals
- Avoiding excessive slicers
- Using appropriate aggregations
- Using `DISTINCTCOUNT` for order-level calculations
- Avoiding incorrect many-to-many relationships
- Testing visual interactions
- Maintaining a clean and simple Executive Overview

Special attention was given to the warehouse operating cost calculation because the operating cost proxy is repeated across order records. Directly summing this field would result in an incorrect inflated value.

Therefore, warehouse-level aggregation was used to calculate the operating cost correctly.

4.10 Final Testing and Validation

The final Power BI report was tested to ensure that the dashboards and navigation worked correctly.

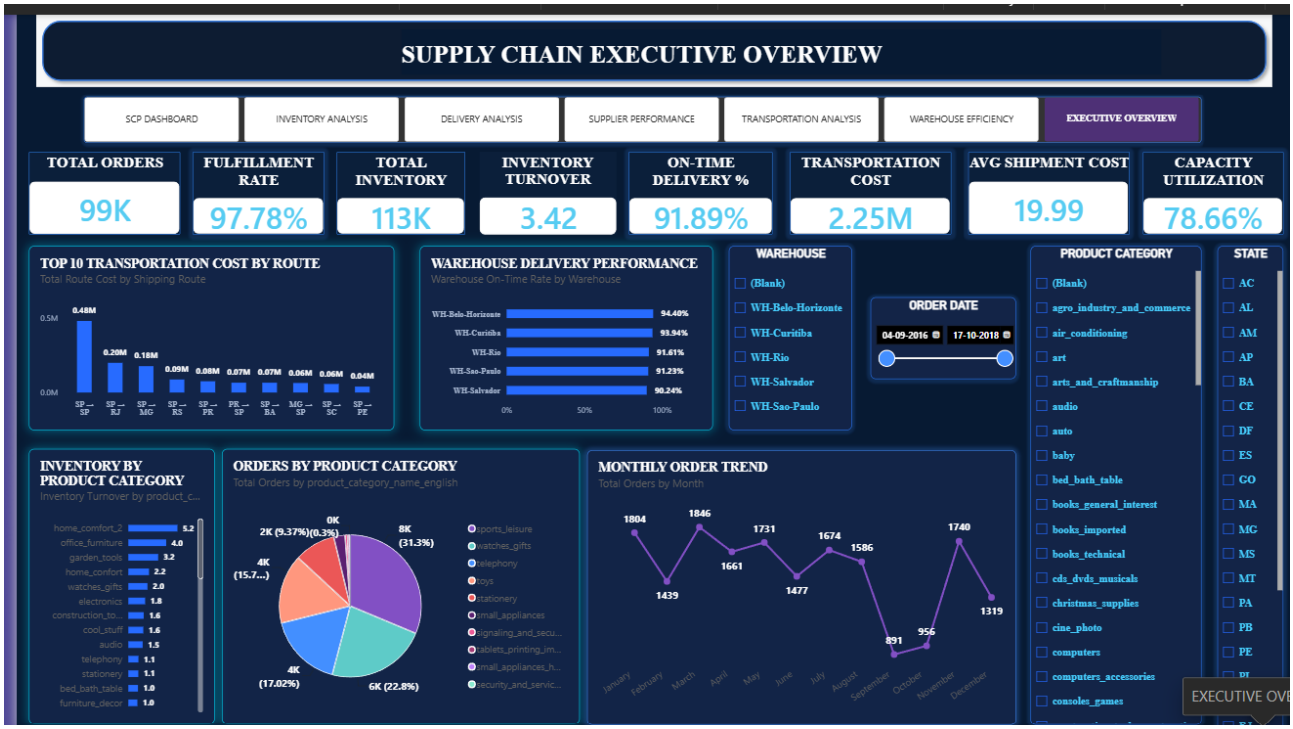
The following areas were validated:

- Executive Overview KPI cards
- Warehouse KPIs
- Warehouse slicers
- Date filters
- Warehouse filters
- Navigation buttons
- Page Navigator
- Home button
- Back button
- Warehouse drill-through
- Transportation Route analysis
- Percentage formatting
- Currency formatting
- Visual interactions
- Dashboard page consistency
- Availability of all five proxy warehouses

The final testing ensured that the report was ready for presentation and deployment.

Dashboard Screenshots

2.1 Executive Overview Dashboard



The Executive Overview Dashboard provides a consolidated view of the overall supply chain performance. It contains major KPIs related to orders, inventory, delivery, transportation and warehouse operations.

The dashboard is designed for management-level users who require a quick understanding of overall supply chain performance without navigating through multiple detailed reports.

Key elements visible in the dashboard include:

- Total Orders
- Fulfillment Rate
- Inventory Performance
- On-Time Delivery
- Transportation Cost
- Warehouse Operating Cost
- Order Trends
- Warehouse Delivery Performance
- Transportation Cost by Route
- Product Category Analysis

2.2 Warehouse Efficiency Dashboard



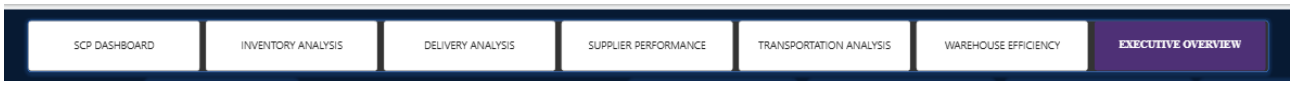
The Warehouse Efficiency Dashboard provides detailed analysis of warehouse performance.

The dashboard compares the proxy warehouses based on:

- Order volume
- Fulfillment rate
- On-time delivery
- Capacity utilization
- Picking accuracy
- Processing time
- Operating cost

This dashboard helps users identify differences in warehouse performance and determine which areas require further investigation.

2.3 Dashboard Navigation



The dashboard navigation system provides buttons for moving between different report pages.

The navigation includes:

Inventory → Delivery → Supplier → Transportation → Warehouse → Executive Overview

The navigation improves the overall user experience and makes the Power BI report easier to operate.

2.4 Navigation Between Dashboard Pages

Issue

The report contains multiple analytical pages, making it difficult for users to move between pages using the default Power BI page tabs.

Resolution

A Page Navigator was implemented using:

Insert → Buttons → Navigator → Page Navigator

This provides a consistent navigation experience across the report.

Milestone 4 – Achievement Summary

- Developed **Warehouse Efficiency Analytics** using proxy warehouse data.
- Created KPIs for **fulfillment, on-time delivery, capacity utilization, picking accuracy, and processing time**.
- Built an **Executive Overview Dashboard** combining inventory, delivery, supplier, transportation, and warehouse insights.
- Implemented **dashboard navigation and warehouse drill-through** functionality.
- Completed **performance optimization, testing, and final dashboard validation**.

CONCLUSION

- The project successfully developed a complete **Supply Chain and Logistics Analytics solution using Power BI** across four milestones.
- **Milestone 1** established the data model, data preparation and core supply chain KPIs.
- **Milestone 2** focused on inventory and delivery performance analysis.
- **Milestone 3** developed supplier and transportation analytics, including route-level analysis.
- **Milestone 4** completed the project with warehouse efficiency analysis, the Executive Overview dashboard, navigation and final testing.
- The final dashboard provides an interactive view of important areas such as **orders, inventory, delivery, suppliers, transportation and warehouse performance**.
- The project also demonstrates the importance of documenting **data limitations and assumptions**, particularly for the proxy warehouse analysis.
- Overall, the project shows how Power BI, Power Query and DAX can convert raw e-commerce data into a **structured, interactive and decision-support dashboard for supply chain management**.

REFERENCES

You can use these additional references in your documentation:

1. **Guy in a Cube** – Power BI tutorials, dashboard design and DAX concepts. https://guyinacube.com/
2. **SQLBI** – DAX, Power BI data modelling and performance optimization resources. https://www.sqlbi.com/
3. **Enterprise DNA** – Power BI dashboard development, visualization and business intelligence resources. https://forum.enterprisedna.co/
4. **RADACAD** – Power BI, Power Query, data modelling and analytics tutorials. https://radacad.com/
5. **DataCamp – Power BI Resources** – Learning resources related to Power BI, data visualization and business intelligence. https://www.datacamp.com/
6. **Kaggle – Olist Brazilian E-Commerce Dataset** – Dataset reference used for the project. https://www.kaggle.com/
7. **Towards Data Science** – Articles and resources related to data analytics, visualization and business intelligence. https://towardsdatascience.com/